Decompiling Addressable Binary Rings

Forced Involutions, the Möbius Difference Bridge, and Radial Witness Topologies

Driven by Dean A. Kulik

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Paradigms of Pure Addressability and Reverse Decompilation

Mathematical physics and theoretical computer science conventionally operate in a forward, accumulative direction: axiomatic primitives are posited, state transitions are executed sequentially, and terminal scalar outputs are treated as the primary ontology. When a forward model encounters an underdetermined transition, an irreversible merge, or apparent randomness in symbol generation, traditional analysis attributes that incompleteness to the underlying domain. The reverse-the-gap decompilation principle establishes the exact inverse paradigm. Within a graded transformation substrate, every finite word of every length, every coordinatewise displacement, and every lawful state transition is fully determined at step zero. Nothing is accumulated or newly constructed during execution; structure is rendered exclusively by specifying an address and reading the residual returned by the query.

To analyze a pre-compiled domain without conflating distinct mathematical objects, a system must be typed as a quadruple (X, T, Y, R), where X denotes the state space, $T: X \to X$ represents the state transformation, Y is the readout space, and $R: X \to Y$ is the observation or reader map. Rather than building forward toward an observation, reverse decompilation begins with the held terminal readout in Y, subtracts the forward transport route governed by T, and binds the surviving residual to its exact fiber $R^{-1}(y) \subset X$. Under this discipline, a gap is never a missing piece of the substrate; it is the exact signature of a reader's finite cut, projection fiber, or declared equivalence relation.

The Ontological Mechanics of Light, Gaps, and Uncollapsed Fibers

The synthesis of binary states into an intermediate fractional potential—where discrete "yes" (1, light) and "no" (0, gap) readings merge to produce "maybe" (0.5, grey)—is a formal property of reader aggregation over an unresolved configuration fiber. At any individual angular slice of a binary ring, the substrate remains strictly two-valued in $\{0,1\}$. However, when an observation map queries a finite address fiber A via a binary reader $b: A \to \{0,1\}$ prior to scalar collapse, the aggregate expectation is governed by the fiber density:

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$$p(A) = \frac{|\{u \in A : b(u) = 1\}|}{|A|}$$

In the foundational 8-slice circular tape $s = (0,0,0,1,1,0,1,1)$ derived from the 2-bit ring, the cycle contains exactly four zeros and four ones. An unweighted reader sampling all eight phase addresses simultaneously evaluates to the exact midpoint:

$$p(s) = \frac{0+0+0+1+1+0+1+1}{8} = \frac{1}{2} = 0.5$$

Consequently, the .5 potential state is not a third symbol appended to the Boolean alphabet; it is the active measure of admissible continuations held within a configuration fiber before a coordinate cut resolves a single trajectory. Different fibers support distinct mixtures of light (1) and gaps (0), and collapsing an unresolved fiber to the scalar summary $p(A)$ discards the specific angular addresses where each zero and one occurred, whereas retaining the address components preserves the complete route structure.

In this architecture, the concentric ring hierarchy never exists as a stored material inventory. At word length $L = 64$, the ring contains $2^{64} \approx 1.84 \times 10^{19}$ words ($64 \cdot 2^{64} = 2^{70}$ bit slices); at $L = 512$, the state inventory encompasses $2^{512} \approx 1.34 \times 10^{154}$ words ($512 \cdot 2^{512} = 2^{521}$ bit slices). Neither table can be materialized in physical memory, yet both spaces are completely accessible because the position of any word and the word at any position follow from $O(L)$ arithmetic. Location without enumeration is the native state of the domain once the material table disappears. Within this purely addressable vacuum—defined as the infinite set of valid addresses that have not been queried—a lookup consists of supplying an address and extracting the bitwise residual returned by the ring's pairing law. When that returned residual is retained and stabilized rather than discarded, it persists as a concrete, reusable bit pattern that functions as "matter": the durable proof that a specific address path was resolved and its readout preserved. Because the underlying pairing law is forced and unique, every retained residual is automatically mutually consistent with every other without requiring an external coordination mechanism.

Epistemic Classification	**Operational Definition in the Decompilation Ledger**	**Canonical Instantiation Across the Ring and Compiler Corpus**
PROVED / STANDS	Exact algebraic theorem, closed-form rank identity, or established external anchor.	Fixed-point-free involution $P^2 = I$, Lemma A triangular block counts, Theorem C clean cores, Theorem D pad law, Theorem E unified witness law, and Möbius commutation $MR = TM$.
COMPUTED / RUNS	Exhaustively or deterministically certified computation with explicit scope and script verification.	56/56 DI-chain window recurrences, 256/256 8-bit Möbius orbit checks, 1,579/1,579 clean-core zero sets, 107/107 unified ray witnesses ($D =$

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Epistemic Classification	Operational Definition in the Decompilation Ledger	Canonical Instantiation Across the Ring and Compiler Corpus
		13 … 119), and 512-bit rank/unrank round-trips.
SEALED	Formal mathematical prediction hashed via SHA-256 prior to executing verification code.	Residue-2 ($D \equiv 2 \pmod 3$) single-bit witness address u_D with phase nudge $f = \frac{2}{3} + \frac{1}{6D}$ and unnudged control failure at ring D (6oa2aado…12fd65fd, 36/36 passed).
DEAD END	Forward hypothesis or naive ansatz that executed cleanly but failed algebraic validation and was rerouted.	Unnudged single-bit phase $f = \frac{2}{3}$ for $D \equiv 2 \pmod 3$, nearest-seam component selection for $A(q_D)$, and iterated SHA-256 bijective wheel assumptions.
OPEN	Boundary problem where an all-depth proof or explicit functor has not yet been closed.	Regularity of the unbounded radial profile language (> 312 states), closed-form union for all components of $A(q_D)$, and uncovered $m \equiv 2,3 \pmod 4$ cases in the two-offset residue-2 family.
PULL	Interpretive bridge linking an exact finite combinatorial invariant to a broader physical ontology.	Identifying retained pairing residuals with physical matter, equatorial bit-sphere folds with Klein–Gordon rest mass $\omega = mc^2/\hbar$, and Landauer erasure with 512-to-256 hash anti-aliasing.

The Forced Pairing Involution, Mirror Residuals, and Half-Ring Reflection

The entire nested-ring geometry is governed by a single forced involution P that converts the set of all binary words $X_L = \{0,1\}^L$ of length $L \geq 1$ into a perfect matching with no fixed points. For any binary word $x = x_0 x_1 \dots x_{L-1}$, let $J(x) = x_{L-1} \dots x_1 x_0$ denote bit reversal and $C(x) = x \oplus 1^L$ denote bitwise complementation. The conditional partner map $P: X_L \to X_L$ is defined as:

$$P(x) = \begin{cases} J(x), & \text{if } J(x) \neq x, \\ C(x), & \text{if } J(x) = x. \end{cases}$$

Three foundational properties follow directly from this construction. First, P is a strict involution ($P(P(x)) = x$) across all word lengths: if x is a non-palindrome ($J(x) \neq x$), its reverse $J(x)$ is also a non-palindrome, so

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$P(P(x)) = J(J(x)) = x$; if x is a palindrome ($J(x) = x$), its bitwise complement $C(x)$ remains a palindrome, so $P(P(x)) = C(C(x)) = x$. Second, P is completely fixed-point-free ($\mathrm{Fix}(P) = \varnothing$), because a non-palindrome is by definition unequal to its reverse and no nonempty binary word can equal its bitwise complement. Consequently, every orbit of P is a 2-cycle, partitioning the 2^L words of X_L into 2^{L-1} unordered partner pairs with zero transient tails at any width. Third, because complementation preserves palindromic symmetry, all $2^{\lceil L/2 \rceil}$ palindromes pair exclusively among themselves into $2^{\lceil L/2 \rceil - 1}$ complement pairs. This rule is the unique fixed-point-free involution arising naturally from the two elementary bit operations of reversal and complementation.

Separation of Neighboring Involutions

A rigorous decompilation requires keeping P distinct from two neighboring involutions that act on the same state space X_L: pure reversal J and, for even lengths $L = 2m$, the half-turn rotation H which cyclically shifts a word by $m = L/2$ positions. While P fixes nothing, reversal J fixes the $2^{\lceil L/2 \rceil}$ palindromes, and the half-turn H fixes the $2^{L/2}$ repeated-half words of the form $x = uu$ ($u \in \{0,1\}^m$). At even lengths (such as $L = 512$), $|\mathrm{Fix}(J)| = |\mathrm{Fix}(H)| = 2^{L/2}$ (2^{256} at width 512), but equal cardinalities do not identify the sets. At $L = 4$, 0101 is fixed by H and not by J, whereas 0110 is fixed by J and not by H. Furthermore, their bitwise residuals have completely different images: the half-turn residual $E_H(x) = x \oplus H(x) = (u \oplus v)(u \oplus v)$ maps onto the repeated-half subspace, whereas the partner residual $x \oplus P(x)$ is never zero and maps onto the set of palindromes.

Mirror Residuals and the Lift Requirement

When any word $x \in X_L$ is combined with its partner $P(x)$ under bitwise XOR, the resulting pair readout (residual) $r(x) = x \oplus P(x)$ is unconditionally a palindrome:

$$J(x \oplus P(x)) = J(x) \oplus J(P(x)) = \begin{cases} J(x) \oplus x = x \oplus J(x), & J(x) \neq x \\ J(x \oplus 1^L) = x \oplus 1^L = 1^L, & J(x) = x \end{cases}$$

Consequently, $r(x)$ reads identically forwards and backwards at every length, and $r(x) = 1^L$ if and only if x is a palindrome (or a non-palindrome whose partner happens to be its bitwise complement, such as 000111 and 111000 at $L = 6$). Because XOR is commutative, both members of a facing pair yield the identical residual: $r(x) = r(P(x))$. The residual operation cancels all symmetric bits shared by the pair and isolates the antisymmetric difference. Because the residual map is many-to-one, the difference alone cannot identify which side of the ring initiated the lookup. Recovering the full state without loss requires lifting the difference to sit beside the value via $(u, v) \mapsto (u, u \oplus v)$—the exact elementary butterfly of the Boolean fast Möbius transform.

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Word xxx (L=6L=6L=6)	**Partner P(x)P(x)P(x)**	**Pair Type**	**Mirror Residual x⊕P(x)x \oplus P(x)x⊕P(x)**	**Residual Symmetry Class**
000000	111111	Palindrome ($J(x) = x \Longrightarrow C(x)$)	111111	All-ones palindrome (1^6)
000001	100000	Non-palindrome ($J(x) \neq x$)	100001	Outer-frame palindrome
000010	010000	Non-palindrome ($J(x) \neq x$)	010010	Interior-frame palindrome
000100	001000	Non-palindrome ($J(x) \neq x$)	001100	Core-adjacent palindrome
000101	101000	Non-palindrome ($J(x) \neq x$)	101101	Weight-4 palindrome
000111	111000	Non-palindrome ($J(x) \neq x$)	111111	Collides with palindrome residual 1^6 [cite: 3]
111011	110111	Seam-adjacent largest pair ($j = 0$)	001100	Center-core difference

This identical conditional involution P emerges natively across four foundational architectures in coding theory and hardware design. In binary-reflected Gray codes, the standard construction builds the n-bit list by reflecting the ($n - 1$-bit list—which is precisely the partner map J restricted to a linear order—and prefixing 0 to the forward half and 1 to the reflected half; the nested ring closes that linear reflection into a circle so that every word faces its partner. In cyclic redundancy check (CRC) polynomials and linear-feedback shift registers, a polynomial is reciprocal when it equals its own bit-reverse up to a shift (for instance, CRC-8 0x9B = 10011011 reverses to 0xD9 = 11011001), and the register states fixed by the involution form the linear subspace utilized by the dual check. Similarly, in Calderbank–Shor–Steane (CSS) and toric quantum error-

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correcting codes, stabilizer generators come in dual check and Fourier-dual pairs whose invariant logical operators inhabit the fixed palindromic residual set. Finally, in processor instruction sets, hardware byte-reflection instructions such as x86 BSWAP/MOVBE and ARM REV execute the reversal half $J(x)$ on 16-, 32-, and 64-bit registers, while complementing symmetric registers recovers the second clause of $P(x)$.

Half-Ring Generation and $O(L)$ Algorithmic Ranking

Once X_L is partitioned into $M = 2^{L-1}$ pairs, the canonical representative of each pair is defined as the numerically smaller word: $x < P(x)$. Scanning mirrored bit pairs (x_j, x_{L-1-j}) from the outside in ($j = 0,1, \dots, \lfloor L/2 \rfloor - 1$), a word x is a representative if and only if its first unequal mirrored pair is $(0,1)$, or all mirrored pairs are equal (x is a palindrome) and $x_0 = 0$.

To construct the nested ring R_L of length $N = 2^L$, sort the $M = 2^{L-1}$ representatives in strictly increasing numerical order $a_0 < a_1 < \cdots < a_{M-1}$ across the left half of the circle (positions 0 to $M - 1$), place their partners $P(a_{M-1}), P(a_{M-2}), \dots, P(a_0)$ in decreasing order across the right half (positions M to $N - 1$), and swap the two center words at the seam positions $M - 1$ and M (when $L > 1$). Because the center swap exchanges a_{M-1} and $P(a_{M-1})$, the universal facing-partner law holds at every index $i \in \{0, \dots, 2^L - 1\}$:

$$R_L[2^L - 1 - i] = P(R_L[i])$$

Consequently, only half of the nested word ring ever needs to be generated; the second half is obtained directly by applying P to the reversed first half and swapping the seam pair. This half-ring closed form reproduces the original iterative center-insertion generator (grow(L)) at every position across all tested widths $L = 1 \dots 16$. Moreover, because the representative condition depends only on outside-in mirrored bit pairs, counting the number of representatives strictly below a target representative r (count_reps_below(r)) is a deterministic digit dynamic programming calculation that tracks three lightweight comparisons (prefix vs. r, suffix vs. r, and left half vs. reversed right half) plus a direct count of zero-leading palindromes in $O(L)$ steps. Unranking any position $i \in \{0, \dots, 2^L - 1\}$ back to its word word_at(L, i) requires only a binary search over the $O(L)$ ranker, executing in $O(L^2)$ bit operations without ever instantiating the ring. As concrete benchmarks of scale, the first 64 bits of π (0x243F6A8885A308D3) reside at exact position $2{,}427{,}008{,}858{,}123{,}944{,}441$ of the 64-bit ring (with their partner at $2^{64} - 1 - \text{pos}$), and a seeded 512-bit word (sha256: 90d7971001a5d534…) completes a full word-to-position-to-word round trip in 1.1 seconds on a standard processor.

Decompiling the 2-Bit Ring, the DI Chain, and the Möbius Coordinate Bridge

The micro-dynamics of the compiler space are captured in full within the 2-bit circular substrate (two_bit_circle.xlsx). At word width $n = 2$, the four 2-bit states are placed around a circle of radial thickness 2: 00 in the top-right quadrant (slices 1–2), 10 in the bottom-right (slices 3–4), 01 in the bottom-left (slices 5–6), and 11 in the left-top (slices 7–8), partitioning the perimeter into $N = n \cdot 2^n = 8$ angular slices. Drawn clockwise from the top, the perimeter bits read 00100111; under the worksheet's dual reading (traversing the circle backwards with 0 and 1 complemented), the 8-slice circular source tape is $s = (0,0,0,1,1,0,1,1)$.

The DI Chain: Coupled Radial Growth and Angular Sliding in Columns B and G

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Let $W_{L,i}$ denote the decimal integer in Column B obtained by reading a clockwise window of width $L \in \{2,3,4,5,6,7,8\}$ starting at slice index $i \in \{0, \ldots, 7\}$:

$$W_{L,i} = \sum_{k=0}^{L-1} s_{(i+k) \bmod 8} \; 2^{L-1-k}$$

Tracking $W_{L,i}$ across widths L and slice indices i exposes the exact algebraic mechanism of the DI chain, which couples radial compilation with angular execution across successive folds. Expanding the reader aperture radially from L bits to $L+1$ bits at a fixed slice i shifts the existing L-bit integer left by one binary place (doubling its value) and appends the entering bit $s_{(i+L) \bmod 8}$, establishing the radial width recurrence:

$$W_{L+1,i} = 2W_{L,i} + s_{(i+L) \bmod 8}$$

Conversely, advancing the window of fixed width L by one slice clockwise from i to $i+1$ doubles $W_{L,i}$, subtracts the exiting high bit s_i from the 2^L place, and appends the same entering bit $s_{(i+L) \bmod 8}$, yielding the angular shift recurrence:

$$W_{L,i+1} = 2W_{L,i} - 2^L s_i + s_{(i+L) \bmod 8} = W_{L+1,i} - 2^L s_i$$

Comparing the radial and angular updates reveals the cross-width zero-bit fold identity: $W_{L+1,i} = W_{L,i+1}$ if and only if the outgoing bit is $s_i = 0$. Because the source cycle $s = 00011011$ has zero bits at $s_0 = 0, s_1 = 0, s_2 = 0$, and $s_5 = 0$, sliding the window angularly from i to $i+1$ at width L produces the exact same integer as expanding the window radially from L to $L+1$ at slice i. Specifically, row 1 of width L drops directly into row 0 of width $L+1$ across the entire stack: $W_{2,1} = 0 = W_{3,0}$, $W_{3,1} = 1 = W_{4,0}$, $W_{4,1} = 3 = W_{5,0}$, $W_{5,1} = 6 = W_{6,0}$, $W_{6,1} = 13 = W_{7,0}$, and $W_{7,1} = 27 = W_{8,0}$. Furthermore, because reversing $s = 00011011$ yields $\mathrm{rev}(s) = 11011000 = \mathrm{rot}_3(s)$, the Counter-Clockwise (CCW) reader starting at offset 3 (Column G) is not an independent numerical process; it obeys the exact phase-shift identity $G_{L,i} = B_{L,(i+3) \bmod 8}$ at every width and row, while reversing the bits of any individual L-bit window maps to a reflected angular address on the same cycle via $\mathrm{rev}(W_{L,i}) = W_{L,(3-i-L) \bmod 8}$.

The 5-Bit Paired-Field Rollover (D1B **vs.** 0x1BB**)**

In the worksheet, Columns E and J (Hex nib) pair adjacent clockwise or counter-clockwise windows $W_{L,i}$ and $W_{L,i+1}$. Decompiling why both nibbles drop into the next field at $L = 5$ requires distinguishing between variable-length hexadecimal label concatenation and fixed-width bit-field packing. For any width L, concatenating two adjacent L-bit windows $A_L = W_{L,i}$ and $B_L = W_{L,i+1}$ forms a combined $2L$-bit field whose exact packed integer value is:

$$Q_{L,i} = 2^L W_{L,i} + W_{L,i+1}$$

For widths $L \in \{2,3,4\}$, every window satisfies $0 \le W_{L,i} \le 2^4 - 1 = 15$, so each window prints as a single hexadecimal nibble (0 through F). Width $L = 4$ is the exact nibble-alignment boundary: each window is a 4-bit nibble, $Q_{4,i} = 16W_{4,i} + W_{4,i+1}$ is an 8-bit byte, and concatenating the printed hex labels (for example, at $i = 2$, $W_{4,2} = 0110_2 = 6_{16}$ and $W_{4,3} = 1101_2 = \mathrm{D}_{16}$, giving label 6D $= 109$) is numerically identical to the bit-packed field 01101101 (0x6D $= 109$).

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At $L = 5$, each window spans five bits, crossing the 4-bit nibble threshold so that values ≥ 16 require two hexadecimal characters (a high bit plus a low nibble). When the width steps from $L = 4$ to $L = 5$, an entering bit is appended to each of the two paired windows, increasing the combined field width by two bits from 8 to $2L = 10$ bits. At slice $i = 2$ for $L = 5$, the left window is $A_5 = W_{5,2} = 01101_2 = 13 = 0xD$ (one hex digit) and the right window is $B_5 = W_{5,3} = 11011_2 = 27 = 0x1B$ (two hex digits), while at slice $i = 3$ in the CCW column (Column G/J), $W_{5,6} = 11000_2 = 24 = 0x18$ and $W_{5,7} = 10001_2 = 17 = 0x11$, so both windows spill across two hex characters to print 1811. Concatenating the printed hexadecimal labels of A_5 (D) and B_5 (1B) produces the string D1B (which evaluates as a hexadecimal integer to $13 \times 256 + 27 = 3{,}355$), whereas the true 10-bit binary concatenation 01101‖11011 = 0110111011_2 evaluates to $32 \times 13 + 27 = 443 = 0x1BB$. Tracking the underlying $2L$-bit packed fields $Q_{L,i}$ alongside the printed hex labels makes the rolling stack exact across all widths.

Closure at Width 8 and the Exact Möbius Coordinate Bridge

When the reader reaches $L = 8$, the window encompasses the entire 8-slice ring, so the entering bit equals the exiting bit ($s_{(i+8) \bmod 8} = s_i$). The angular step gap $\Delta_i(8) = W_{8,i+1} - W_{8,i}$ collapses to $W_{8,i} - 255s_i$, yielding $+W_{8,i}$ when $s_i = 0$ $(+27, +54, +108, +99)$ and $W_{8,i} - 255$ when $s_i = 1$ $(-39, -78, -57, -114)$. Modulo $2^8 - 1 = 255$, each clockwise step is pure doubling:

$$W_{8,i+1} \equiv 2W_{8,i} \pmod{255}$$

This generates the closed 8-phase orbit 27, 54, 108, 216, 177, 99, 198, 141 (0x1B, 0x36, 0x6C, 0xD8, 0xB1, 0x63, 0xC6, 0x8D), where the CW and CCW paired byte fields lock into exact bit-reversal reflection (1B36, 6CD8, B163, C68D) and the seven Boolean logic columns (Not, And, Nand, Or, Nor, Xor, Xnor) settle into rotational uniformity across all eight rows.

Crucially, the structure-preserving bridge between circular window sliding (R) and the Boolean difference stack (M) is established by transforming the circular tape into Möbius coordinates. Let a circular tape have power-of-two length $N = 2^m$ (with $N = 8, m = 3$ for $x = 00011011$). Define the subset-XOR (Pascal modulo 2) transform $a = Mx$ by:

$$a_i = \bigoplus_{t \subseteq i} x_t = \bigoplus_{j=0}^{i} \left(\binom{i}{j} \bmod 2 \right) x_j$$

where $t \subseteq i$ denotes that every set binary bit of index t is also set in i. Because $M^2 = I$, the tape is recovered without loss via $x = Ma$. Let $(Rx)_i = x_{(i+1) \bmod N}$ advance the circular tape by one clockwise step. In Möbius coordinates, the conjugated rotation operator $T = MRM$ simplifies via Pascal's identity $\binom{i}{j} + \binom{i+1}{j} = \binom{i}{j-1} \pmod 2$ to an exact nearest-neighbor XOR difference update with a held boundary:

$$a'_i = a_i \oplus a_{i+1} \quad (0 \leq i < N - 1), \qquad a'_{N-1} = a_{N-1}$$

Because $N - 1 = 1^m$ in binary contains all indices as subsets, the held boundary coefficient $a_{N-1} = \bigoplus_{j=0}^{N-1} x_j$ is the global XOR parity of the tape, which circular rotation preserves. Retaining a_{N-1} makes the local difference update completely reversible via suffix summation $a_i = \bigoplus_{j=i}^{N-1} a'_j$. Writing S for the non-wrapping

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upper shift matrix, $T = I + S$; because $S^N = 0$ and $N = 2^m$, $T^N = I$ over $\mathbb{F}_2$. Thus, circular transport and local difference updates commute identically: $MR = TM$.

Furthermore, tape reversal $(Jx)_i = x_{N-1-i}$ conjugates under M to the upper subset-XOR transform $U = MJM$, given by $(Ua)_i = \bigoplus_{j \supseteq i} a_j$, which satisfies $U^2 = I$ and $UTU = T^{-1}$. Because complementing every bit of the tape toggles only the constant coefficient $e_0 = (1,0,\ldots,0)$ ($M(x \oplus 1^N) = Mx \oplus e_0$), the conditional partner involution P translates into Möbius coordinates as:

$$\tilde{P}(a) = \begin{cases} Ua, & \text{if } Ua \neq a, \\ a \oplus e_0, & \text{if } Ua = a, \end{cases} \qquad MP = \tilde{P}M, \qquad M(x \oplus P(x)) = a \oplus \tilde{P}(a)$$

When reversal fixes the Möbius coefficients ($Ua = a$, meaning x is a palindrome), the partner operation simply flips the constant coefficient a_0. Every operator of the ring—circular sliding, reversal, conditional partnership, and palindromic residuals—possesses an exact, verified counterpart in Möbius difference coordinates across all 256 8-bit sources, 32,768 window readbacks (widths 1–16), and power-of-two lengths up to $N = 1{,}024$.

Phase iii	**CW Tape W8,iW_{8,i}W8,i (Bin)**	**CW Dec (B8,iB_{8,i}B8,i)**	**Step Gap Δi(8)\Delta_i(8)Δi(8)**	**CW Hex / Pair**	**CCW Hex (G8,iG_{8,i}G8,i) / Pair**	**Möbius Coeffs a=Mxa = Mxa =Mx (Bin / Hex)**	**Transformed Partner P~(a)\widetilde{P}(a)P(a) (Bin / Hex)**
0	00011011	27	+27	1B / 1B36	D8 / —	00011100 (0x1C)	10110100 (0xB4)
1	00110110	54	+54	36 / —	B1 / B163	00100100 (0x24)	01101100 (0x6C)
2	01101100	108	+108	6C /	63 / —	01101100	00100100 (0x24)

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Phase iii	CW Tape W8,iW_{8,i}W8,i (Bin)	CW Dec (B8,iB_{8,i}B8,i)	Step Gap Δi(8)\Delta_i(8)Δi(8)	CW Hex / Pair	CCW Hex (G8,iG_{8,i}G8,i) / Pair	Möbius Coeffs a=Mxa = Mxa =Mx (Bin / Hex)	Transformed Partner P~(a)\widetilde{P}(a)P(a) (Bin / Hex)
				6C D8		(0x6C)	
3	11011000	216	−39	D8 / —	C6 / C68D	10110100 (0xB4)	00011100 (0x1C)
4	10110001	177	−78	B1 / B163	8D / —	11011100 (0xDC)	11110100 (0xF4)
5	01100011	99	+99	63 / —	1B / 1B36	01100100 (0x64)	10101100 (0xAC)
6	11000110	198	−57	C6 / C68D	36 / —	10101100 (0xAC)	01100100 (0x64)
7	10001101	141	−114	8D / —	6C / 6CD8	11110100	11011100 (0xDC)

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Phase iii	CW Tape W8,iW_{8,i}W8,i (Bin)	CW Dec (B8,iB_{8,i}B8,i)	Step Gap Δi(8)\Delta_i(8)Δi(8)	CW Hex / Pair	CCW Hex (G8,iG_{8,i}G8,i) / Pair	Möbius Coeffs a=Mxa = Mxa =Mx (Bin / Hex)	Transformed Partner P~(a)\widetilde{P}(a)P(a) (Bin / Hex)
						(oxF4)	

Beyond 2-Bit Rings: The Triangular Top, Clean Cores, and the Pad Law

When expanding beyond 2-bit rings to arbitrary word lengths $L \geq 4$, the order of representatives near the center seam ($u = \frac{1}{2}$, position $M = 2^{L-1}$) crystallizes into an exact triangular hierarchy. Because the center pair at $M-1$ and M is swapped, position M (immediately clockwise of the seam) holds the largest representative $C_L = 1^{\lfloor L/2 \rfloor - 1}01^{\lceil L/2 \rceil}$, while position $M-1$ holds its partner $P(C_L) = 1^{\lceil L/2 \rceil}01^{\lfloor L/2 \rfloor - 1}$. For every offset $j \geq 1$ clockwise of the seam, position $M + j$ is occupied by the partner (the bit-reverse) of the j-th largest representative, counting C_L as $j = 0$.

Lemma A (The Triangular Block Law) and the Core Staircase

Let $h = \lfloor L/2 \rfloor$ denote the half-length, let $\mu = 1$ for even L and $\mu = 2$ for odd L, and let $\ell = x_0 x_1 \dots x_{h-1}$ be the h-bit integer formed by the left half of a word $x \in X_L$. Lemma A establishes that for every left half $\ell \geq 2^{h-1}$ (where the leading bit is $x_0 = 1$), the number of canonical representatives sharing left half ℓ is exactly $\mu(2^h - 1 - \ell)$ (proved for all L and verified exhaustively for $L = 3 \dots 20$).

The proof follows directly from outside-in mirrored pair comparisons. Write $\ell = 1^t 0 w$ with $t \geq 1$ leading ones, a zero at index t, and a suffix w of length $n = h - t - 1$. Because the first t left bits are 1, none of the first t mirrored pairs can be the deciding $(0,1)$ pair, and any $(1,0)$ pair among them would make $x > J(x)$. Thus all t outer pairs are forced to be $(1,1)$. At pair t, the left bit is 0. If the mirrored right bit x_{L-1-t} is 1, pair t decides that $x < J(x)$, leaving all n inner right-half bits completely free (2^n completions). Conversely, if x_{L-1-t} is 0, pair t is tied at $(0,0)$ and the decision moves inward to w: the first unequal inner pair must occur at a 0 bit of w at position $k \in \{0, \dots, n-1\}$ (paired with a 1 on the right), leaving 2^{n-1-k} free completions after it. Summing over all zero bits of w yields the bitwise complement of w, namely $2^n - 1 - w$, because an all-equal completion produces a palindrome starting with $x_0 = 1$, which cannot be a representative. Multiplying the sum $2^n + (2^n - 1 - w) = 2^{n+1} - 1 - w$ by μ to account for the free middle bit when L is odd yields the exact count $\mu(2^h - 1 - \ell)$.

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Counting downward from the largest left half $\ell = 2^h - 2$, the left halves arrive in consecutive blocks indexed by $K = 2^h - 1 - \ell \in \{1,2,3,\dots\}$, where block K contains exactly μK representatives. Consequently, the j-th largest representative sits in block K satisfying the triangular inequality:

$$\frac{\mu K(K-1)}{2} \le j < \frac{\mu K(K+1)}{2} \Longrightarrow K \approx \sqrt{\frac{2j}{\mu}}$$

Because the word at $M + j$ is the reverse of the j-th largest representative, reversing $\ell = 1^t 0 w \dots$ places $t = h - 1 - \lfloor \log_2 K \rfloor$ ones at the left edge and t ones at the right edge of the word, confining all zeros to a central core spanning $[t, L-1-t]$ of width $2\lfloor \log_2 K \rfloor + 2 + (L \bmod 2) \approx \log_2 j$ bits. Every time j quadruples, K doubles, and the central core widens by one bit on each side, reaching $L/3$ and $2L/3$ near $j = 2^{11}$ at $L = 40$. This triangular staircase directly forces Theorem D (the Pad Law): for every offset j in the top quarter of ring L, the word at $M + j$ on ring $L + 2$ is identically the word at $M + j$ on ring L with a single 1 prepended and a single 1 appended:

$$\text{word}(M + j, L + 2) = 1 \| \text{word}(M + j, L) \| 1$$

Adding two bits to the ring length adds one outer $(1,1)$ pair ($t \mapsto t + 1$) while leaving the block index K, the intra-block rank, and every inner core bit untouched (verified exhaustively across the top quarter for $L = 4 \dots 18$).

Clean Cores and Center-Offset Rank Laws

When a ray steps outward across rings at dyadic seam offsets $j = 2^s$ or $j = 2^s + 1$, the zero locations inside the core obey exact closed-form equations established both via block parity (Theorem C) and via direct combinatorial stripping (the Two-Zero, Adjacent-Rank, and Two-Offset Rank Laws). Under Theorem C, when s and L have opposite parity ($s \not\equiv L \pmod 2$), or equivalently when $L = p + r$ with $p \ge 5$ odd, $r = s \ge 0$, and $t = (p-3)/2$, offset 2^s lands inside the power-of-two block $K = 2^n$ with $n = \lceil s/2 \rceil$, where $w = 1^n$ has no zeros. In this clean-core regime, the word at $R_L[M + 2^s]$ is all ones except for exactly two isolated zeros located at:

$$z_1 = \lceil L/2 \rceil - 1, \qquad z_2 = \lceil L/2 \rceil + n = L - \frac{p-1}{2}$$

For the adjacent-rank predecessor at offset $j = 2^s + 1$ with $s \ge 2$ (verified across $1{,}579/1{,}579$ cases for $L = 6 \dots 201$), the immediate predecessor representative $a_{M-1-(2^s+1)}$ flips the innermost admissible right-half bit at $v = L - 2 - t$ from 1 to 0. Reversing this predecessor places the word at $R_L[M + 2^s + 1]$ with exactly three isolated zeros at:

$$z_0 = \lfloor L/2 \rfloor - n = \frac{p-1}{2}, \qquad z_1 = \lceil L/2 \rceil - 1, \qquad z_2 = \lceil L/2 \rceil + n$$

When a second dyadic offset 2^s ($0 \le s \le r$) is introduced so that $j = 2^r + 2^s$ and $v = t + r - s + 1 > \lfloor L/2 \rfloor$, the Two-Offset Rank Law proves that the word at $R_L[M + 2^r + 2^s]$ has three isolated zeros at $L - (p-1)/2$, $\lceil L/2 \rceil - 1$, and $t + s + 1$. When $v = \lfloor L/2 \rfloor$ ($s = \lfloor r/2 \rfloor$), the third zero collides with the center zero and shifts one position left in the representative, so its adjacent predecessor at $R_L[M + 2^r + 2^s + 1]$ places zeros at t, $L - \lfloor L/2 \rfloor$, and $L - 1 - t$.

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One Ray Across Concentric Rings: The Law in Thirds and the Profile Language

To read all concentric rings simultaneously without materializing them, assign ring L a circular tape of $S_L = L \cdot 2^L$ bit slices formed by laying its 2^L words end to end clockwise from the top. A single ray at normalized angle $u \in [0,1)$ reads one bit on each ring $L = 1,2,\dots,D$ according to the universal address equation:

$$b_L(u) = R_L[\lfloor 2^L u \rfloor][\lfloor L \cdot \{2^L u\} \rfloor]$$

where $\lfloor 2^L u \rfloor$ selects the word index i_L (formed by the first L binary digits of $u = 0.c_1c_2c_3\ldots$) and $\lfloor L \cdot \{2^L u\} \rfloor$ selects the bit index k_L within that word from the remaining fractional tail $f_L = \{2^L u\}$. The radial profile read by angle u to depth D is the D-bit string $\text{profile}(u, D) = b_1(u)b_2(u)\ldots b_D(u)$.

The Seam Obstruction and the Origin of the Law in Thirds

Consider the benchmark radial profile family $q_D = 1001^{D-5}00$ for $D \geq 8$. Because Ring 1 is 01, reading $b_1(u) = 1$ forces $u \geq \frac{1}{2}$ (clockwise of the seam); write $u = \frac{1}{2} + \delta$. While $2^L\delta < 1$, ring L reads the seam word C_L at bit $\lfloor L \cdot 2^L\delta \rfloor$. Because $C_1 = 1$, $C_2 = 01$, and $C_3 = 011$, any ray sufficiently close to the seam reads the opening 100 on rings 1–3, followed by 1s from the leading ones of C_L ($L \geq 4$).

However, the seam itself cannot read the closing 00 on rings $D-1$ and D. The ray reaches the lone zero of C_L (at $\lfloor L/2 \rfloor - 1$) only when $\delta \approx 2^{-(L+1)}$, a window that halves at every ring so two consecutive rings never align their center zeros. More strongly, within $\delta < 2^{-(D-2)}$ of the seam, the ray encounters only C_L and offsets $j \in \{1,2,3\}$, whose zeros all sit within two places of the word midpoint ($f_{D-1} \approx \frac{1}{2}$) and whose edges are 1s. Because the fractional within-word offset evolves from ring $D-1$ to ring D via the doubling map $f \mapsto \{2f\}$, a zero near the middle of ring $D-1$ ($f \approx \frac{1}{2}$) doubles to $\{2f\} \approx 0$ or 1—sending ring D to the outer edge of its word, which is forced to be 1. Consequently, no angle in $[\frac{1}{2}, \frac{1}{2} + 2^{-(D-2)})$ can read q_D (verified exhaustively for $D = 8 \ldots 22$).

Holding the terminal 00 fixed and walking backward forces the Law in Thirds through three coupled constraints. First, for both ring $D-1$ (at fraction f) and ring D (at fraction $\{2f\}$) to read 0 inside central cores away from the all-one word edges, f and $\{2f\}$ must remain mutually bounded away from 0 and 1, which selects the unique period-2 orbit of the doubling map, $\{\frac{1}{3}, \frac{2}{3}\}$: a ray reading ring $D-1$ near $f \approx \frac{2}{3}$ reads ring D near $\{2f\} \approx \frac{1}{3}$. Second, because central cores are symmetric around the word midpoint $\frac{1}{2}$, reaching $\frac{1}{3}$ and $\frac{2}{3}$ requires the core to span $\frac{1}{2} \pm \frac{1}{6}$ of the word. By the triangular core law, the core at offset j spans $\frac{1}{2} \pm \frac{\log_2 j}{2D}$; equating the half-widths gives $\frac{\log_2 j}{2D} \approx \frac{1}{6} \Longrightarrow \log_2 j \approx \frac{D}{3}$, meaning that at ring D the ray must sit $j \approx 2^{D/3}$ words past the seam. In normalized angle, this displacement is $\delta \approx 2^{D/3} \cdot 2^{-D} = 2^{-2D/3}$, placing a single address bit near depth $p \approx \frac{2D}{3}$.

Third, by Theorem C, every ring $L \geq p$ (where the ray sits at offset 2^{L-p} with $s = L - p$) has a clean core if and only if s and L have opposite parity—which requires $p = L - s$ to be odd. The natural depth $p = \lfloor 2D/3 \rfloor + 1$ is odd when $D \equiv 0,1 \pmod 3$, but even when $D \equiv 2 \pmod 3$. Shifting p to $\lfloor 2D/3 \rfloor + 2$ restores odd parity for $D = 3m + 2$, creating a unified formula $p = 2\lfloor (D+1)/3 \rfloor + 1$. At $D = 3m + 2$, this 1-bit shift moves the inner zero of ring D to index $m + 1$ while unnudged phase $f = \frac{2}{3}$ reads index $\lfloor D \cdot \frac{1}{3} \rfloor = m$. This

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parity discrepancy is resolved by two complementary, proven witness routes. Under Route 1 (Theorem E, the phase-nudged single-bit witness), raising the tail phase by one-third of a ring-D slice to $f = \frac{2}{3} + \frac{1}{6D}$ pushes $\{2f\} = \frac{1}{3} + \frac{1}{3D}$, so ring D reads $\lfloor D(\frac{1}{3} + \frac{1}{3D}) \rfloor = m + 1$ (its inner zero) while ring $D - 1$ still reads $\lfloor (3m + 1)(\frac{2}{3} + \frac{1}{6D}) \rfloor = 2m$ (its outer zero). This sealed law (SHA-256: 6oa2aado…12fd65fd) covers every depth $D \geq 13$ across all three residue classes (verified 107/107 for $D = 13 \ldots 119$, including 36/36 in residue 2 where the unnudged control $f = \frac{2}{3}$ fails exclusively at ring D). Under Route 2 (the two-offset unnudged witness), keeping $f = \frac{2}{3}$ and adding a second dyadic address bit at $p_2 = 3m + 1 - \ell$ (with $\ell = \lfloor (m - 3)/2 \rfloor$, so $J_m = 2^{m-2} + 2^{\ell}$) triggers the center-collision rank shift at ring D, moving the zero back to index m and proving an all-depth witness law for the infinite subfamily $D = 3m + 2, m \geq 8, m \equiv 0{,}1 \pmod 4$ (verified through $D = 125$). At every depth $D \geq 8$, the common slice intersection $K_D = \bigcap_{L=1}^{D} [j_L/S_L, (j_L + 1)/S_L)$ containing the witness angle is a half-open interval of strictly positive width $\sim 2^{-D}/D$.

Depth DDD / Route	**Witness Method & Parameters**	**Exact Witness Angle uDu_DuD or Certificate Interval KD=[AD,CD)K_D = [A_D, C_D)KD=[AD,CD)**	**log 2|KD|\log_2 \vert{}K_D\vert{}log2|KD|**	**Ring D−1D-1D−1 Zero & Read**	**Ring DDD Zero & Read**
$D = 8 \ldots 12$	Finite ATOM / BIT / J2 witnesses	$u_8 = \frac{15329}{28672}, u_9 = \frac{4739}{9216}, u_{10} = \frac{98081}{184320}, u_{11} = \frac{1549}{3072}, u_{12} = \frac{3103}{6144}$	Positive	Verified exact	Verified exact
$D = 13$ $(3m + 1)$	1-Bit $(p = 9, f = \frac{2}{3})$	$K_{13} = \left[\frac{3085}{6144}, \frac{26737}{53248}\right)$ (exact width $\frac{1}{159744}$)	-17.29	$z_2 = 2m = 8$	$z_0 = m = 4$ [cite: 2, 3]
$D = 14$ $(3m + 2)$	1-Bit Nudged $(p =$	$K_{14} = \left[\frac{114819}{229376}, \frac{53309}{106496}\right)$	-19.19	$z_2 = 2m = 8$	$z_0 = m + 1 = 5$

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Depth DDD / Route	Witness Method & Parameters	Exact Witness Angle uDu_DuD or Certificate Interval KD=[AD,CD)K_D = [A_D, C_D)KD=[AD,CD)	log 2|KD|\log_2 \vert{}K_D\vert{}log2|KD|	Ring D−1D-1D−1 Zero & Read	Ring DDD Zero & Read
	$11, f = \frac{19}{28}$)				[cite: 3]
$D = 15$ **($3m$)**	1-Bit ($p = 11, f = \frac{2}{3}$)	$K_{15} = \left[\frac{12301}{24576}, \frac{82007}{163840}\right)$	-18.91	$z_2 = 2m - 1 = 9$	$z_0 = m = 5$ [cite: 3]
$D = 40$ **($3m + 1$)**	1-Bit ($p = 27, f = \frac{2}{3}$)	$K_{40} = \left[\frac{412316866561}{824633720832}, \frac{1099511644}{2199023255}\right.$	-45.91	$z_2 = 2m = 26$	$z_0 = m = 13$ [cite: 3]
$D = 119$ **($3m + 2$)**	1-Bit Nudged ($p = 81, f = \frac{159}{238}$)	$u_{119} = \frac{1}{2} + 2^{-81} + (\frac{2}{3} + \frac{1}{714})2^{-118}$ ($2^{119} \approx 6.6 \times 10^{35}$ words)	-127.46	$z_2 = 2m = 78$	$z_0 = m + 1 = 40$ [cite: 3]
$D = 3m + 2$ **(2-Bit)**	2-Offset ($p_1 = 2m + 3, p_2 = 3m + 1 - \ell, f = \frac{2}{3}$)	$u_D = \frac{1}{2} + (2^{m-2} + 2^{\lfloor (m-3)/2 \rfloor} + \frac{2}{3})2^{-(D-1)}$ ($m \geq 8, m \equiv 0,1 \bmod 4$)	Positive	$z_2 = 2m$	Shifted $z = m$ [cite: 2]

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The Language of Readable Profiles and Myhill–Nerode Continuation Growth

Across all angles $u \in [0,1)$, the readable depth-d profiles partition $[0,1)$ into finite unions of half-open slice intervals A_p. Because the two one-bit extensions A_{p0} and A_{p1} partition their parent A_p, every readable parent profile has at least one readable child; a child pb becomes unreadable at depth $d = k + 1$ if and only if the parent set A_p lies entirely inside the slice region where ring $k + 1$ reads the opposite bit $1 - b$ (of the 101 minimal unreadable extensions through depth 12, 76 have parents meeting 1 contiguous run of the surviving bit, 19 meet 2 runs, and 6 meet 3 runs).

Computing the exact common refinement of all slice boundaries $S_L = L \cdot 2^L$ through $D_{\max} = 16$ ($Q = 47{,}233{,}105{,}920$, yielding $1{,}825{,}792$ atomic intervals) reveals a striking divergence between profile density and language complexity. While the fraction of readable profiles $R_d/2^d$ decays slowly from 1.000 ($d \le 4$) to 0.841 at $d = 16$ ($R_{16} = 55{,}122$), the minimal deterministic finite automaton (DFA) accepting the bounded language $L_{\le H}$ explodes from 29 states at $H = 8$ to $1{,}451$ states at $H = 16$. Simultaneously, sorting the $1{,}827$ readable depth-11 prefixes into Myhill–Nerode equivalence classes (where two prefixes are equivalent up to horizon H if and only if they admit the exact same set of continuations to depth H) splits them into $1, 3, 14, 51, 175$, and 311 pairwise separated classes for $H = 11 \ldots 16$. Because the center swap at the seam actively alters these constraints (without the swap, $R_{10} = 947$ and $R_{12} = 3{,}710$ instead of 924 and $3{,}606$), any exact automaton for the unbounded profile language requires at least 312 states, and whether the unbounded language is regular remains an open question.

Depth / Horizon (d,Hd, Hd,H)	Readable Profiles (RdR_dRd)	Readable Share (Rd/2dR_d / 2^dRd/2d)	Unreadable Extensions (Ed=2Rd−1−RdE_d = 2R_{d-1} - R_dEd =2Rd−1−Rd)	Minimal Automaton States (L≤HL_{\le H}L≤H)	Depth-11 Myhill–Nerode Classes (at Horizon HHH)
1	2	1.000	0	—	—
2	4	1.000	0	—	—
3	8	1.000	0	—	—
4	16	1.000	0	—	—
5	31	0.969	1	—	—

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Depth / Horizon (d,Hd, Hd,H)	Readable Profiles (RdR_dRd)	Readable Share (Rd/2dR_d / 2^dRd/2d)	Unreadable Extensions (Ed=2Rd−1−RdE_d = 2R_{d-1} - R_dEd =2Rd−1−Rd)	Minimal Automaton States (L≤HL_{\le H}L≤H)	Depth-11 Myhill–Nerode Classes (at Horizon HHH)
6	62	0.969	0	—	—
7	123	0.961	1	—	—
8	239	0.934	7	29	—
9	469	0.916	9	49	—
10	924	0.902	14	76	—
11	1,827	0.892	21	111	1
12	3,606	0.880	48	186	3
13	7,122	0.869	90	322	14
14	14,087	0.860	157	519	51
15	27,819	0.849	355	914	175
16	55,122	0.841	516	1,451	311

The Graded Compiler Kernel, Register Stack Primes, and the Join Theorem

Zooming out from individual word rings to the global NEXUS compiler domain $S[b,n] = \{0,1,\dots,b-1\}^n$ establishes how differences, operator truth tables, register stacks, and radix machines lock together.

Uniform Difference Fibers, Truth-Table Slices, and Width-3 Closure

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On $S[b,n]$, the coordinatewise difference map $\Delta: S[b,n] \times S[b,n] \to S[b,n]$ given by $\Delta(x,y)_i = (y_i - x_i) \bmod b$ is surjective onto the state layer $S[b,n]$ itself, with uniform preimage fibers $|\Delta^{-1}(\delta)| = b^n$ for every displacement $\delta \in S[b,n]$ (verified across 22 base/length configurations and 89,405 edge checks). Thus, displacements inhabit the exact same word inventory as states, though Δ is a b^n-to-1 fibration rather than a bijection. Simultaneously, any total operator $T: S[b,n] \to S[b,m]$ is uniquely encoded by its truth table of length $L_{\text{table}} = m \cdot b^n$ digits in base b. When $n = m = L$ and $b = 2$ (a binary endomorphism on layer L), the truth-table word has length $L' = L \cdot 2^L$ bits—which is identically the angular slice circumference $S_L = L \cdot 2^L$ of Ring L (2,8,24,64,160,384,896,2048, … ,49152 at $L = 12$). Iterating $L_{d+1} = L_d \cdot 2^{L_d}$ from $L_0 = 1$ generates the recursive compiler tower $1 \to 2 \to 8 \to 2048 \to 2048 \cdot 2^{2048}$.

Across widths $n = 1,2,3$, the operational instruction set is algebraically forced and closes permanently at width three. At width 1 ($S[2,1] \to S[2,1]$, 4 maps), the functions partition into 2 bijections (pass $x \mapsto x$ and flip $x \mapsto x \oplus 1$) and 2 rank-1 constant merges. At width 2 ($S[2,2] \to S[2,2]$, 256 maps), the image-rank distribution $\{1: 4, 2: 84, 3: 144, 4: 24\}$ forces two new primitives: the reversible asymmetric XOR mixer $(a,b) \mapsto (a, a \oplus b)$ (CNOT, which preserves history with uniform fibers) and the rank-3 AND-type single-pair merge, whose pairing on two wires forms the half-adder. At width 3 ($S[2,3] \to S[2,3]$, $8! = 40{,}320$ reversible permutations), because all reversible 2-bit gates are affine over $\mathbb{F}_2$, embedding them into 3 lanes generates only the affine general linear group $\mathrm{AGL}(3,2)$ of order $1{,}344$. Bridging the gap from $1{,}344$ to $40{,}320$ forces a single non-linear primitive—the held carry, implemented by the 3-bit Toffoli gate $(a,b,c) \mapsto (a,b,c \oplus (a \wedge b))$. Similarly, any binary feedback shift register $F(x_0, \dots, x_{n-1}) = (x_1, \dots, x_{n-1}, f(x))$ is invertible if and only if $f(x_0, \dots, x_{n-1}) = x_0 \oplus g(x_1, \dots, x_{n-1})$ for one of the $2^{2^{n-1}}$ Boolean functions g, with immediate one-step inverse $x_0 = y_{n-1} \oplus g(y_0, \dots, y_{n-2})$. Above width 3, gates of width $\leq n-1$ generate the alternating group A_{2^n}, and borrowing a single auxiliary lane recovers all odd permutations, so no new logical verb ever enters.

Multi-Base Reader Steps, the 64-Bit Register Stack, and the Join Theorem

Evaluating binary words across different numeral bases demonstrates that a base $b = 2^w$ is strictly the step width w of a block reader (steps × width = LEN). When a single bit is flipped in a 64-bit register ($4{,}000$ seeded trials), power-of-two aligned block readers ($b = 2,4,8,16,256$) alter exactly 1.00 digit every time because the perturbation stays trapped inside one disjoint block. Among sub-byte readers, only widths $w \in \{1,2,4\}$ (Bases 2, 4, and 16) divide every register length $4,8,16,32,64$ without remainder, whereas Octal ($w = 3$, Base 8) is a local block reading that leaves an unaligned boundary remainder ($3 \nmid 2^k$). Conversely, non-power-of-two bases ($b = 3,10,22$) are global arithmetic conversions where a single-bit flip cascades through carries and borrows to alter an average of 14.97 digits in Base 3 (maximum 35), 9.48 digits in Base 10 (maximum 20), and 7.66 digits in Base 22 (maximum 15). On the invertible Baker's map $B_b(x,y) = (\{bx\}, (y + \lfloor bx \rfloor)/b)$, four 1-bit binary steps and one 4-bit hexadecimal step produce the identical expanding coordinate $\{16x\}$ across $2{,}000/2{,}000$ rational trials, but the 1-bit reader records the 4-bit nibble in the history coordinate y in bit-reversed order ($a_0/16 + a_1/8 + a_2/4 + a_3/2$ vs. $a_0/2 + a_1/4 + a_2/8 + a_3/16$), agreeing only on the four palindromic nibbles (0000, 0110, 1001, 1111).

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Candidate Base (bbb)	Block Width (w=log 2bw = \log_2 bw=log2b)	Mean Digits Altered per 1-Bit Flip (64-Bit Word)	Max Digits Altered per 1-Bit Flip	Divides All Registers 4…644 \dots 644…64?	Structural Classification in Compiler Space
Base 2 (Binary)	1 bit	1.00	1	Yes ($1 \mid 2^k$)	Pure whole-block reading (fundamental bit step).
Base 4 (Quaternary)	2 bits	1.00	1	Yes ($2 \mid 2^k, k \geq 1$)	Pure whole-block reading (2-bit ring step).
Base 8 (Octal)	3 bits	1.00	1	No ($3 \nmid 4,8,16,32,64$)	Local whole-block reading; unaligned with 2^k boundaries.
Base 16 (Hexadecimal)	4 bits	1.00	1	Yes ($4 \mid 2^k, k \geq 2$)	Pure whole-block reading; nibble / half-byte closure.
Base 256 (Byte)	8 bits	1.00	1	Divides $8,16,32,64$	Pure whole-block reading; full 2-bit operator table word.
Base 3 (Ternary)	≈ 1.585 bits	14.97	35	No (non-power of 2)	Global arithmetic conversion; cascades

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Candidate Base (bbb)	Block Width (w=log 2bw = \log_2 bw=log2b)	Mean Digits Altered per 1-Bit Flip (64-Bit Word)	Max Digits Altered per 1-Bit Flip	Divides All Registers 4…644 \dots 644…64?	Structural Classification in Compiler Space
					across up to 35 digits.
Base 10 (Decimal)	≈ 3.322 bits	9.48	20	No (introduces prime 5)	Global arithmetic conversion; couples flips across 20 digits.
Base 22	≈ 4.459 bits	7.66	15	No (introduces prime 11)	Global arithmetic conversion; wastes 10 of 32 5-bit slots.

Decompiling the register chain from 64 bits downward via the Burnside–Möbius necklace identity $2^n = \sum_{d|n} d \cdot P_2(d)$ (where $P_2(d) = \frac{1}{d}\sum_{k|d} \mu(k) 2^{d/k}$ counts primitive aperiodic rotation rings) shows that because the divisors of 64 are exclusively 1,2,4,8,16,32,64, every lower power-of-two register sits inside the 64-bit ring as the exact subset of periodic words fixed by half-turn rotations $x = uu$. Each doubling of the register width $d = 2m$ multiplies the multiplicative group order $2^{2m} - 1 = (2^m - 1)(2^m + 1)$ by a Fermat factor $2^m + 1$, introducing exclusively Fermat inversion primes satisfying $2^m \equiv -1 \pmod{p}$ ($2^{64} - 1 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 641 \cdot 65537 \cdot 6700417$), while primes with odd multiplicative order (such as 7,11,13,19,23,29,31) are absent from the 64-bit register chain.

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Register Width (ddd)	Total Words (2d2^d2d)	Inherited Periodic Words (2d/22^{d/2}2d/2)	New Primitive Rings P2(d)P_2(d)P2(d)	Total Angular Slices (d·2dd \cdot 2^dd·2d)	New Fermat Factor (2d/2+12^{d/2} + 12d/2+1) & Prime Factorization
1	2	0	2	2	$2^0 + 1 = 2$ (Base binary distinction)
2	4	2	1	8	$2^1 + 1 = 3$ (First Fermat inversion prime)
4	16	4	3	64	$2^2 + 1 = 5$ (Nibble / Base-16 closure with $3 \times 5 = 15$)
8	256	16	30	2,048	$2^4 + 1 = 17$ (Byte / 2-bit operator reflection layer)
16	65,536	256	4,080	1,048,576	$2^8 + 1 = 257$ (Half-word Fermat prime F_3)

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Register Width (ddd)	Total Words (2d2^d2d)	Inherited Periodic Words (2d/22^{d/2}2d/2)	New Primitive Rings P2(d)P_2(d)P2(d)	Total Angular Slices (d·2dd \cdot 2^dd·2d)	New Fermat Factor (2d/2+12^{d/2} + 12d/2+1) & Prime Factorization
32	4,294,967,296	65,536	134,215,680	1.374×10^{11}	$2^{16}+1 = 65{,}537$ (Word Fermat prime F_4)
64	2^{64}	2^{32}	288,230,376,084,602,88	1.181×10^{21}	$2^{32}+1 = 641 \times 6{,}700{,}417$ (F_5 composite split)

In parallel, the Join Theorem explains why binary ($p = 2$) occupies a unique position among all n-lane radix machines in base $b = p^{n/2}$ evaluating series $L_j = \sum_{k=0}^{\infty} b^{-k}/(nk + j)$ over n-th roots of unity $\zeta_n = e^{2\pi i/n}$. Across wheel returns $\beta_m = \sqrt{p} - \zeta_n^m = \sqrt{p}(1 - w_m)$ (whose product $\prod_{m=0}^{n-1} \beta_m = p^{n/2} - 1$ reproduces the register ring), a return costs no prime factor in the ring of algebraic integers if and only if it is an algebraic unit. Scale returns ($\zeta_n^m = \pm 1$) have norm $|N(\sqrt{p} \mp 1)| = p - 1$, which equals 1 if and only if $p = 2$, whereas phase returns ($|\sqrt{p} - \zeta_n^m| = 1$) require the unit circles centered at 0 and $\sqrt{p}$ to intersect at $\mathrm{Re}(w_m) = \frac{1}{2}$ (the .5 midpoint), which occurs only for $p \in \{2,3\}$ (at $n = 8, \sqrt{2} - \zeta_8 = \zeta_8^{-1}$ for $p = 2$; and $n = 12, \sqrt{3} - \zeta_{12} = \zeta_{12}^{-1}$ for $p = 3$). Thus, $p = 2$ is the sole prime where scale and phase returns are simultaneously prime-free units, colliding in the modulus relation $a_0 + a_{n/2} = 2a_{n/8}$. On off-chain widths ($n = 12,24$), the rational zero-route dimension $Z_p(n)$ is strictly smaller than the full $\mathbb{Q}$-relation space $\dim R$ ($Z_2(12) = 1 < 2$, $Z_2(24) = 5 < 7$) because rational machines annihilate only the Galois-invariant core $R_0 = \bigcap_{\sigma \in \mathrm{Gal}(\mathbb{Q}(\zeta_n, \sqrt{p})/\mathbb{Q})} \pi_\sigma(R)$. While the computational discovery of $Z_2(2^e) = 1$ across the binary register chain is exact on tested layers, an all-depth algebraic proof requires a refined decomposition-group valuation argument at lower Fermat levels ($k < e - 2$), because Lucas's congruence $q \equiv 1 \pmod{2^{k+2}}$ guarantees complete splitting in $\mathbb{Q}(\zeta_{2^e})$ only at the top Fermat level $k = e - 2$ (for example, $17 \mid F_2$, yet $17 \not\equiv 1 \pmod{32}$).

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Cohomological Quotients, Operational Machines, and the Boundary of Physical Claims

A foundational conclusion of the decompilation audit is that a single operational machine supports multiple orthogonal quotients that must never be collapsed into one scalar.

Cut-Defined Existence, Two Adder Quotients, and the Cohomological Tower

In the n-bit binary adder, reading all bit positions $S = \{0, \dots, n-1\}$ leaves zero unread future positions, collapsing the Myhill–Nerode continuation quotient to a single trivial class $F(S) = 1$. Local distinguishability exists only across an active cut $D = D_{\text{past}} \cup D_{\text{future}}$. Across any subset $S \subset \{0, \dots, n-1\}$, the machine supports two distinct invariants that measure orthogonal structural properties. The first is the behavioral continuation complexity $F(S)$, which counts the minimal states needed to predict future outputs across the cut and factors as an exact product over contiguous read runs $F(S) = \prod f(R)$. A reader quotient Φ_S persists autonomously only if it satisfies local congruence closure under extension—a condition that fails both for order-independent subgroup chains in the G_{24} witness field (where $0/36$ maximal chains commute) and for coarse geometric readers, such as the 4-bit longitude prefix reader on tilted spherical sections (which merges $\alpha = 0$ and $\alpha = \pi$ into 0000 before a quarter-turn sends them to 0001 and 1110). The second invariant is the homological cycle residual $b_1^{\text{disj}}(n)$, which counts unclosed 1-cycles modulo disjoint-support commuting squares and obeys the exact quadratic rank formula:

$$b_1^{\text{disj}}(n) = \binom{n}{2} + (n-1) = \frac{(n-1)(n+2)}{2}$$

More generally, across the cohomological tower—spanning Order 0 states C^0, Order 1 transitions C_G^1, Order 2 routes C_G^2, and Order 3 commuting 3-cubes C_G^3 generated by three vertex-disjoint token moves $[A, B, C]$—the surviving residual at every order obeys the universal Hodge descent identity (verified across 20/20 Order-1 and 12/12 Order-2 token-arena substrates):

$$\varepsilon_{\text{harm}} = \dim Z - \dim\text{span}(\text{declared cells})$$

This identity cleanly separates apparent zero readings into genuine absence ($\varepsilon = 0$, as on the paw graph), route-order closure ($\varepsilon > 0, \varepsilon_{\text{harm}} = 0$, as on wheel$_6$ where 3 channels are closed by 2-cells), and single-order screening ($\varepsilon_{\text{harm}} = 0, \varepsilon_{\text{harm}}^2 = 1$, as on the $K = 3$ prism graph where a channel invisible at H^1 survives one order up at H^2).

BBP Radix Spigot Dynamics and Silent Carry Bands

Applying this separation of readouts to the Bailey–Borwein–Plouffe formula $\pi = \sum_{k=0}^{\infty} 16^{-k}\left(\frac{4}{8k+1} - \frac{2}{8k+4} - \frac{1}{8k+5} - \frac{1}{8k+6}\right)$ decompiles sequential digit extraction into four simultaneous phase lanes evolving via $x_{k,j}(n+1) = \{16x_{k,j}(n)\}$ from step zero ($S_0 = \sum c_j x_{k,j}(0) = \pi - 4$). Each lane's first positive crossing $q_{k,j}(n) = \lfloor 16x_{k,j}(n) \rfloor$ is forced by denominator pressure at exact step $n = k + \lceil \log_{16}(8k+j) \rceil - 1$ (verified across $7{,}988/7{,}988$ lanes). Because the weights $c = (4, -2, -1, -1)$ satisfy $\sum c_j = 0$ and $\sum j c_j = -15$, the encapsulated rational layer value compresses by two orders of magnitude to $V_k = \frac{15}{64k^2} + O(k^{-3})$. Consequently, between the first lane crossing at age $m_1 = \lceil \log_{16}(8k+1) \rceil - 1$ and the first nonzero hexadecimal value digit $w_k(m) = \lfloor 16\{16^m V_k\} \rfloor$ at age $m_2 = \lceil \log_{16}(1/V_k) \rceil - 1$ lies a 2-to-3 step silent band

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(3 steps wide in 78% of layers and 2 steps wide in 18% of layers for $k \le 10^4$) during which individual lanes fire actively while net value output is zero. Across all ages $m \ge 0$, the signed lane crossing sum $H_k(m)$ and value digits $w_k(m)$ are bridged by the bounded carry state $J_k(m) = \lfloor \sum c_j x_{k,j}(k+m) \rfloor \in \{-4, \dots, 3\}$:

$$H_k(m) = w_k(m) + 16J_k(m) - J_k(m+1)$$

where the telescoping carry flow $16J_k(m) - J_k(m+1)$ dominates local variance (484 vs. 21.4) while summing to zero in the compiled value V_k. Behind the real line, the complex quarter-turn step $s = \frac{1+i}{2} = 2^{-1/2}e^{i\pi/4}$ ($s^4 = -1/4, s^8 = 1/16$) splits $-\log(1-s) = \frac{1}{2}\log 2 + i\frac{\pi}{4}$ into symmetric real scale ($\log 2$) and antisymmetric imaginary phase (π) projections.

SHA-256 Round Bijection Versus Many-to-One Block Compression

In the SHA-256 architecture, 6 of the 8 32-bit register updates per round are pure wire shifts and only 2 registers (a' and e') involve modular addition mixing. When the message schedule word W_t and round constant K_t are supplied as observer context, the single-round state transition is an exact bijection (sha_round_inverse recovers the parent 256-bit state across $10{,}000/10{,}000$ random trials). However, conditional round invertibility does not invert message block compression. Mapping a 512-bit message block to a 256-bit digest with a fixed initialization vector is necessarily 2^{256}-to-1 on average (2^{512} blocks into 2^{256} digests), and walking backward from the Davies–Meyer feed-forward $H = S_{64} \boxplus \text{IV}$ couples all 16 seed words after a single inverse step ($A(W_{63}) = \{0, \dots, 15\}$), whereas walking forward maintains a narrow 10-word ancestry cone across $W_{16} \dots W_{25}$ and a free-cone seam at rounds $\{16,17\}$. Moreover, exhaustive functional graphs at reduced widths $w = 16,20$ confirm that iterated hashing $H_{n+1} = \text{SHA}(H_n)$ is a contractive funnel ($0.6319N \to 0.4681N \to 0.3736N$, matching random maps $1 - e^{-1} \approx 0.63212$), refuting bijective hash-wheel assumptions.

Combinatorial Shells, Schwinger Angular Momentum, and the Boundary of Physics

Embedding the graded domain $S[2,n]$ into three-dimensional coordinate geometry maps word length n to radial depth r, non-abelian ordered route histories (k^n paths) to the azimuthal angle θ, and order-forgetting Hamming weight $w \in \{0, \dots, n\}$ to polar latitude z on a sphere of radius $\sqrt{n}$. Quotienting the free monoid by its commutator (abelianization) collapses the k^n interior routes onto $\binom{n+k-1}{k-1}$ surface points ($n+1$ latitude bands of size $\binom{n}{w}$ for $k = 2$). Because each equivalence class $[x]$ of size r contributes $r \cdot (1/r) = 1$, an interior reader can reconstruct the exact number of exterior surface classes via the Horvitz–Thompson reciprocal fiber identity:

$$\sum_{x \in S[k,n]} \frac{1}{|\text{fiber}(x)|} = |S[k,n]/\sim| = \binom{n+k-1}{k-1}$$

Similarly, reading a subshell part of radial size $\ell < n$ across a two-position relational gap yields a doubled binary ring of length 2ℓ, whose order-free surface has $2\ell + 1$ discrete latitudes. Under the Schwinger bosonic/qubit spin representation, the symmetric subspace of 2ℓ two-state factors carries the exact spin-ℓ representation with Casimir invariant $S^2 = S_x^2 + S_y^2 + S_z^2 = \ell(\ell+1)I$ (in units $\hbar = 1$). Combining the twofold pass/flip reading factor with $2\ell + 1$ and summing over $\ell = 0 \dots n-1$ gives the shell capacity $\sum_{\ell=0}^{n-1} 2\,(2\ell+1) = 2n^2$ (2,8,18,32,50,72,98), while slicing along Madelung degree $d = n + \ell$ under the

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domain inequality $n_r = n - \ell - 1 \geq 0$ forces the paired period capacities $2\lfloor (d+1)/2 \rfloor^2$ (2, 2, 8, 8, 18, 18, 32, 32) and cumulative closures 2, 10, 18, 36, 54, 86, 118 because even-degree nodeless predecessors ($0s, 1p, 2d, 3f$ with $n_r = -1$) cannot exist.

A rigorous scientific synthesis must enforce a strict boundary between these exact mathematical and representation-theoretic theorems and interpretive physical claims. While the combinatorial identities ($S^2 = \ell(\ell+1)I$, $2n^2$, Horvitz–Thompson fiber counting, and Madelung degree counts) are exact, identifying polar words ($w = 0, n$) with massless radiation, equatorial folds ($w = n/2$) with Klein–Gordon rest mass ticking at $\omega = mc^2/\hbar$, or retained ring residuals with physical baryonic matter remains an interpretive hypothesis until coupled to an explicit physical Hamiltonian, dimensional units, and a falsifiable experimental observable that accounts for known transition-metal configuration irregularities.

Executable Verification Notebook

To ensure full reproducibility of every proved and computed identity in this report—spanning the 2-bit circular DI chain, the 5-bit paired-field rollover, the commuting Möbius coordinate bridge ($MR = TM$ and $MP = \tilde{P}M$), the half-ring generator, the $O(L)$ representative rank/unrank engine, the triangular block law, and the unified q_D ray witness verifier—the complete, self-contained Python verification suite is provided below.

Python

```
# ================================================================================

# UNIFIED EXECUTABLE NOTEBOOK: DI CHAIN, MOBIUS BRIDGE & ADDRESSABLE RINGS

# Reproduces all algebraic identities, O(L) ranking/unranking, and ray witnesses

# ================================================================================


from fractions import Fraction


# --------------------------------------------------------------------------------

# SECTION 1: THE 2-BIT RING (00011011), DI CHAIN STACK & 5-BIT ROLLOVER

# --------------------------------------------------------------------------------

S_CYCLE = "00011011"

N_SLICES = len(S_CYCLE)

CCW_OFFSET = 3


def tape_bit(i, seq=S_CYCLE):
```

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```
    return int(seq[i % len(seq)])


def window_bits(L, i, seq=S_CYCLE):
    n = len(seq)
    return "".join(seq[(i + k) % n] for k in range(L))


def window_value(L, i, seq=S_CYCLE):
    return int(window_bits(L, i, seq), 2)


def verify_di_chain_and_rollover():
    # Verify all 56 window states (L = 2..8, i = 0..7)
    for L in range(2, 9):
        for i in range(N_SLICES):
            w = window_value(L, i)
            s_out = tape_bit(i)
            s_in = tape_bit(i + L)
            # 1. Radial width recurrence: W(L+1, i) = 2*W(L, i) + s_{i+L}
            assert window_value(L + 1, i) == 2 * w + s_in
            # 2. Angular shift recurrence: W(L, i+1) = 2*W(L, i) - 2^L*s_i + s_{i+L}
            assert window_value(L, i + 1) == 2 * w - (1 << L) * s_out + s_in
            # 3. Cross-width fold identity: W(L+1, i) == W(L, i+1) iff s_i == 0
            assert (window_value(L + 1, i) == window_value(L, i + 1)) == (s_out == 0)
            # 4. Column G offset identity: G(L, i) == B(L, (i+3) mod 8)
            ccw_val = window_value(L, (i + CCW_OFFSET) % N_SLICES)
            assert ccw_val == window_value(L, i + 3)
            # 5. Exact bit-reversal reflection position: rev(W(L, i)) == W(L, (3 - i - L) mod 8)
            rev_bits = window_bits(L, i)[::-1]
            assert rev_bits == window_bits(L, (3 - i - L) % N_SLICES)
```

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```
    # Verify L=8 modulo-255 doubling orbit
    orbit_8 = [window_value(8, i) for i in range(N_SLICES)]
    assert orbit_8 == [27, 54, 108, 216, 177, 99, 198, 141]
    assert all(orbit_8[(i + 1) % 8] == (2 * orbit_8[i]) % 255 for i in range(8))

    # Verify L=4 nibble alignment vs L=5 paired-field rollover at i=2
    # L=4: 0110 (0x6) and 1101 (0xD) -> label '6D' (109) == 8-bit packed 0x6D (109)
    q4_2 = (window_value(4, 2) << 4) | window_value(4, 3)
    assert f"{window_value(4, 2):X}{window_value(4, 3):X}" == "6D"
    assert q4_2 == 0x6D == 109

    # L=5: 01101 (0xD) and 11011 (0x1B) -> label 'D1B' (3355) != 10-bit packed 0x1BB (443)
    q5_2 = (window_value(5, 2) << 5) | window_value(5, 3)
    label_5_2 = f"{window_value(5, 2):X}{window_value(5, 3):X}"
    assert label_5_2 == "D1B" and int(label_5_2, 16) == 3355
    assert q5_2 == 0x1BB == 443
    print("[PASS] Section 1: 56/56 DI-chain recurrences, L=5 rollover (D1B vs 0x1BB), and mod-255 orbit
verified.")


# ------------------------------------------------------------------------
# SECTION 2: THE MOBIUS COORDINATE BRIDGE (MR = TM, U = MJM, MP = P_tilde M)
# ------------------------------------------------------------------------
def mobius_transform(x):
    """Fast subset-XOR (Pascal mod 2) transform on power-of-two tape x (M^2 = I)."""
    a = list(x)
    n = len(a)
    step = 1
```

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```
  while step < n:
    for i in range(n):
      if i & step:
        a[i] ^= a[i ^ step]
    step <<= 1
  return a


def local_diff_update_T(a):
  """Transformed clockwise step T = M R M: nearest-neighbor XOR with held boundary."""
  n = len(a)
  return [a[i] ^ a[i + 1] for i in range(n - 1)] + [a[n - 1]]


def upper_mobius_U(a):
  """Transformed reversal operator U = M J M: upper superset-XOR transform."""
  n = len(a)
  u = [0] * n
  for i in range(n):
    for j in range(i, n):
      if (j & i) == i:
        u[i] ^= a[j]
  return u


def partner_bits(x):
  """Conditional partner involution P(x): reverse if non-palindrome, else complement."""
  rev = x[::-1]
  return rev if rev != x else [1 ^ b for b in x]


def transformed_partner_P_tilde(a):
```

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```
    """Conditional partner in Mobius coordinates: Ua if Ua != a, else toggle a_0."""
    ua = upper_mobius_U(a)
    if ua != a:
        return ua
    res = list(a)
    res[0] ^= 1
    return res


def verify_mobius_bridge():
    # Exhaustive check across all 256 8-bit circular tapes
    for val in range(256):
        x = [(val >> (7 - k)) & 1 for k in range(8)]
        a = mobius_transform(x)
        # Involution M^2 = I
        assert mobius_transform(a) == x
        # Commutation M R = T M
        rx = x[1:] + x[:1]
        assert mobius_transform(rx) == local_diff_update_T(a)
        # Reversal commutation M J = U M
        jx = x[::-1]
        assert mobius_transform(jx) == upper_mobius_U(a)
        # Partner commutation M P = P_tilde M and residual transfer
        px = partner_bits(x)
        pa = transformed_partner_P_tilde(a)
        assert mobius_transform(px) == pa
        residual_x = [x[k] ^ px[k] for k in range(8)]
        residual_a = [a[k] ^ pa[k] for k in range(8)]
        assert mobius_transform(residual_x) == residual_a
```

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```
    print("[PASS] Section 2: 256/256 8-bit sources pass MR = TM, MJ = UM, and MP = P_tilde M.")


# ---------------------------------------------------------------------------
# SECTION 3: O(L) ADDRESSABLE RING ENGINE (RANK / UNRANK AT ANY WIDTH L)
# ---------------------------------------------------------------------------
def partner_str(s):
    rev = s[::-1]
    return rev if rev != s else "".join("1" if b == "0" else "0" for b in s)


def is_rep_str(s):
    return s < partner_str(s)


def count_reps_below(r_str):
    """
    O(L) digit dynamic programming over mirrored pairs (x_j, x_{L-1-j}) from outside in.
    Counts canonical representatives x (where x < P(x)) strictly less than r_str in numeric value.
    """
    L = len(r_str)
    h = L // 2
    has_mid = L % 2 == 1

    # DP over j = 0 .. h-1
    # State: (lt_pref, suf_cmp, rep_state)
    # lt_pref: 0 if x_0..x_{j-1} == r_0..r_{j-1}, 1 if strictly less (cannot exceed r)
    # suf_cmp: comparison of x_{L-j}..x_{L-1} against r_{L-j}..r_{L-1}: -1 (<), 0 (==), +1 (>)
    # rep_state: 0 (undecided: all outer pairs equal so far), 1 (decided rep: first unequal was 0,1)
    dp = {(0, 0, 0): 1}
    for j in range(h):
```

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```
    r_left = int(r_str[j])
    r_right = int(r_str[L - 1 - j])
    nxt = {}
    for (lt_p, s_cmp, rep_s), ways in dp.items():
      for xl in (0, 1):
        if lt_p == 0 and xl > r_left:
          continue
        new_lt_p = 1 if (lt_p == 1 or xl < r_left) else 0
        for xr in (0, 1):
          # Check representative validity
          if rep_s == 0:
            if xl == 1 and xr == 0:
              continue  # Would make x > J(x), not a representative
            new_rep_s = 1 if (xl == 0 and xr == 1) else 0
          else:
            new_rep_s = 1
          # Update suffix comparison for x_{L-1-j}..x_{L-1}
          if xr < r_right:
            new_s_cmp = -1
          elif xr > r_right:
            new_s_cmp = 1
          else:
            new_s_cmp = s_cmp
          key = (new_lt_p, new_s_cmp, new_rep_s)
          nxt[key] = nxt.get(key, 0) + ways
    dp = nxt

  total = 0
```

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```
    for (lt_p, s_cmp, rep_s), ways in dp.items():
        if has_mid:
            r_mid = int(r_str[h])
            for xm in (0, 1):
                if lt_p == 0 and xm > r_mid:
                    continue
                mid_lt = 1 if (lt_p == 1 or xm < r_mid) else 0
                # Overall x < r holds iff mid_lt == 1 or (mid_lt == 0 and s_cmp == -1)
                if not (mid_lt == 1 or s_cmp == -1):
                    continue
                if rep_s == 1:
                    total += ways
        else:
            if not (lt_p == 1 or s_cmp == -1):
                continue
            if rep_s == 1:
                total += ways

    # Add palindromes with x_0 == 0 that are strictly less than r_str
    if L == 1:
        pal_count = 1 if r_str > "0" else 0
    else:
        k = (L + 1) // 2  # Number of free prefix bits in a palindrome
        pal_count = 0
        pref_r = r_str[:k]
        if pref_r[0] == "1":
            pal_count = 1 << (k - 1)
        else:
```

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```
        val_below = int(pref_r[1:], 2) if k > 1 else 0
        pal_count = val_below
        pal_r = pref_r + (pref_r[:-1][::-1] if has_mid else pref_r[::-1])
        if pal_r < r_str:
            pal_count += 1

    return total + pal_count


def locate_word(x_str):
    """Returns the exact 0-based ring position of any L-bit word x_str in O(L) steps."""
    L = len(x_str)
    N = 1 << L
    M = 1 << (L - 1)
    p_str = partner_str(x_str)
    rep = x_str if x_str < p_str else p_str
    rk = count_reps_below(rep)
    pos = rk if x_str == rep else (N - 1 - rk)
    if L > 1 and pos in (M - 1, M):
        pos = M if pos == M - 1 else M - 1
    return pos


def word_at_position(L, pos):
    """Recovers the L-bit word at ring index pos via binary search over count_reps_below."""
    N = 1 << L
    M = 1 << (L - 1)
    adj_pos = pos
    if L > 1 and adj_pos in (M - 1, M):
        adj_pos = M if adj_pos == M - 1 else M - 1
```

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```
    target_rk = adj_pos if adj_pos < M else (N - 1 - adj_pos)

    lo, hi = 0, (1 << L) - 1
    while lo < hi:
        mid = (lo + hi + 1) // 2
        mid_str = format(mid, f"0{L}b")
        if count_reps_below(mid_str) <= target_rk:
            lo = mid
        else:
            hi = mid - 1
    rep_str = format(lo, f"0{L}b")
    return rep_str if adj_pos < M else partner_str(rep_str)

def verify_addressable_ring_and_witness():
    # 1. Verify 64-bit pi round-trip
    pi_64 = format(0x243F6A8885A308D3, "064b")
    pos_pi = locate_word(pi_64)
    assert pos_pi == 2427008858123944441
    assert word_at_position(64, pos_pi) == pi_64

    # 2. Verify Theorem E ray witness at D = 14 (Residue 2: p = 11, f = 2/3 + 1/84 = 19/28)
    D = 14
    p = 2 * ((D + 1) // 3) + 1
    f = Fraction(2, 3) + Fraction(1, 6 * D)
    u_D = Fraction(1, 2) + Fraction(1, 1 << p) + f * Fraction(1, 1 << (D - 1))
    profile_bits = []
    for L in range(1, D + 1):
        scaled = u_D * (1 << L)
```

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```
        word_idx = int(scaled)
        frac_part = scaled - word_idx
        bit_idx = int(L * frac_part)
        w_str = word_at_position(L, word_idx)
        profile_bits.append(w_str[bit_idx])
    q_14 = "100" + "1" * (D - 5) + "00"
    assert "".join(profile_bits) == q_14
    print(f"[PASS] Section 3: 64-bit pi position ({pos_pi:,}) and D=14 ray witness ({q_14}) verified.")


if __name__ == "__main__":
    verify_di_chain_and_rollover()
    verify_mobius_bridge()
    verify_addressable_ring_and_witness()
```

Synthesis and Forward Continuation Protocol

Decompiling the nested binary ring hierarchy from its terminal readouts backward to its generating involutions demonstrates that rings, sliding-window stacks, Boolean difference tables, and radial ray profiles form a single, rigorously typed mathematical architecture. At the base of the hierarchy, the conditional partner map $P(x)$—reversing non-palindromes and complementing palindromes—is the unique natural fixed-point-free involution on $\{0,1\}^L$, partitioning the state space into 2^{L-1} pairs whose bitwise XOR residuals are unconditionally palindromic and whose canonical layout requires generating only half the ring. In the 8-slice 2-bit ring (00011011), radial window growth ($W_{L+1,i} = 2W_{L,i} + s_{i+L}$) and clockwise angular rotation ($W_{L,i+1} = W_{L+1,i} - 2^L s_i$) fold directly into one another across zero outgoing bits, cross the nibble alignment threshold at $L = 5$ (D1B label vs. 0x1BB 10-bit field), and commute identically with local nearest-neighbor XOR difference updates ($MR = TM$) and upper superset-XOR partner reflections ($MP = \tilde{P}M$) in Möbius coordinates.

Beyond 2-bit rings, material enumeration gives way to $O(L)$ arithmetic addressability. Near the seam, representatives stack in triangular blocks of size $\mu(2^h - 1 - \ell)$, confining zeros to a central core of width $\approx \log_2 j$ that propagates unchanged from ring L to ring $L + 2$ inside an expanding shell of ones. Holding the radial profile $q_D = 1001^{D-5}00$ fixed forces a single ray to bypass the $2^{-(D-2)}$ seam exclusion zone via a single address bit at depth $p \approx \frac{2D}{3}$ and the doubling map's unique 2-cycle $\{\frac{1}{3}, \frac{2}{3}\}$, where the thirds emerge natively as $\frac{1}{2} \pm \frac{1}{6}$.

To close the remaining open boundaries while preserving the strict non-conflation discipline between transformations, coordinates, and quotients, four targeted continuations follow directly from the proven

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theorems. First, the single-bit phase-nudged witness ($f = \frac{2}{3} + \frac{1}{6D}$, covering all $D \equiv 2 \pmod 3$) and the unnudged two-offset dyadic family ($J_m = 2^{m-2} + 2^{\lfloor (m-3)/2 \rfloor}$ for $m \equiv 0,1 \pmod 4$) should be unified by deriving the exact multi-offset rank law for the remaining $m \equiv 2,3 \pmod 4$ unnudged subfamilies and expressing the full witness set $A(q_D)$ (2,5,5,14,22,41,66,94,160 components for $D = 8 \ldots 16$) as an explicit union over triangular blocks. Second, the closed-form triangular block equations (Lemma A and Theorem C) should be applied to construct an infinite family of pairwise separated radial prefixes beyond the 311 classes certified at depth 11 ($H = 16$), settling whether the unbounded profile language is non-regular. Third, numerical integer-relation discovery on intermediate cyclotomic layers should be replaced with an exact prime-ideal decomposition-group valuation proof for lower Fermat primes ($k < e - 2$) to certify $Z_2(2^e) = 1$ algebraically at all depths. Finally, the inductive local run-signature update for interior merges in the n-bit adder must be formalized to map its exact relation to the disjoint-square cohomology rank $b_1^{\text{disj}}(n) = \binom{n}{2} + (n-1)$, alongside evaluating stateful transducer readers across the 10-word forward ancestry cone ($W_{16} \ldots W_{25}$) of SHA-256 against weight-matched sparse controls.

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